Chapter 1
SET THEORY

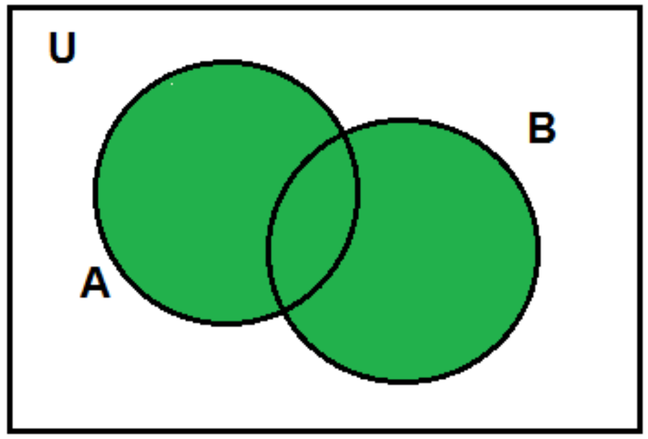
[Part 2: Operation on Set]

Union

- The union of two sets A and B, denoted by $A \cup B$, is defined to be the set

$$A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$$

- The union consists of all elements belonging to either A or B (or both)



Venn diagram of $A \cup B$

Example

$A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6\}$ and $C = \{8, 9\}$

$A \cup B = \{1, 2, 3, 4, 5, 6\}$

$A \cup C = \{1, 2, 3, 4, 5, 8, 9\}$

$B \cup C = \{2, 4, 6, 8, 9\}$

$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 8, 9\}$

Union

If A and B are finite sets, the cardinality of $A \cup B$,

$$| A \cup B| = |A| + |B| - |A \cap B|$$

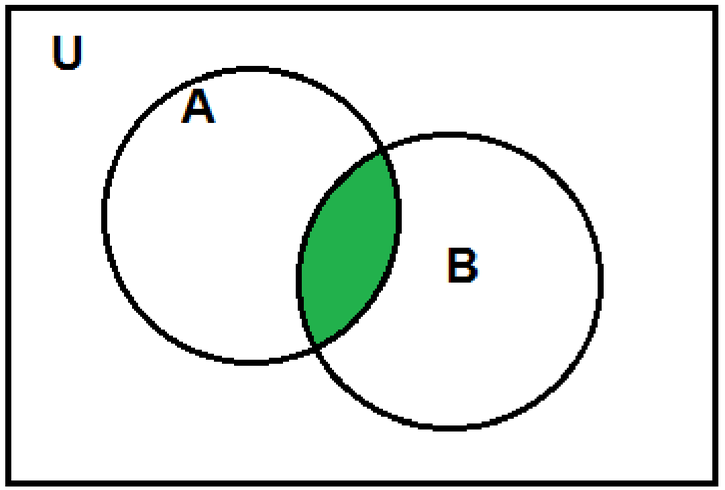
Intersection

- The intersection of two sets A and B, denoted by $A \cap B$, is defined to be the set

$$A \cap B = \{ x \mid x \in A \text{ and } x \in B\}$$

- The intersection consists of all elements belonging to both A and B.

Venn diagram of $A \cap B$



Example

$A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8, 10\}$ and $C = \{ 1, 2, 8, 10 \}$

$A \cap B = \{2, 4, 6\}$

$A \cap C = \{1, 2\}$

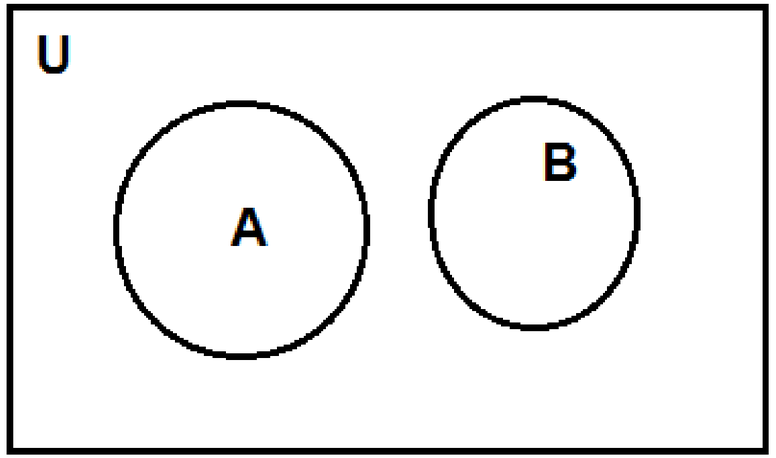
$C \cap B = \{2, 8, 10\}$

$A \cap B \cap C = \{2\}$

Disjoint

Two sets A and B are said to be disjoint if, $A \cap B = \emptyset$

Venn diagram, $A \cap B = \emptyset$



Example

$A = \{1, 3, 5, 7, 9, 11\}$, $B = \{2, 4, 6, 8, 10\}$

$A \cap B = \emptyset$

Difference

The set,

$A-B = \{x \mid x \in A \text{ and } x \notin B\}$ is called the difference.

The difference $A - B$ consists of all elements in A that are not in B.

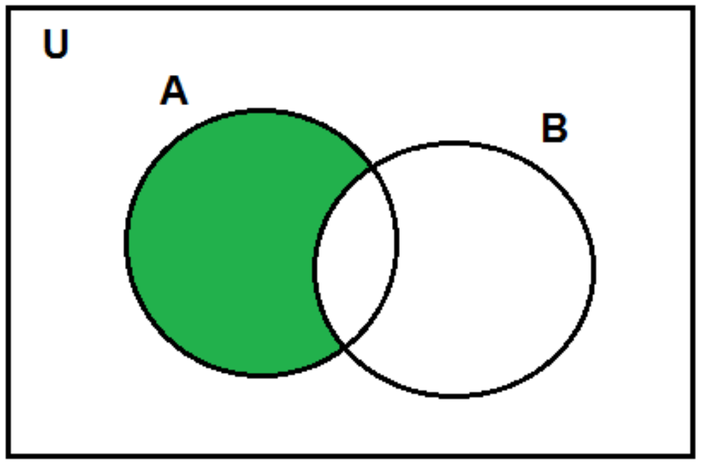
Example

$A = \{ 1, 2, 3, 4, 5, 6, 7, 8 \}$

$B = \{ 2, 4, 6, 8 \}$

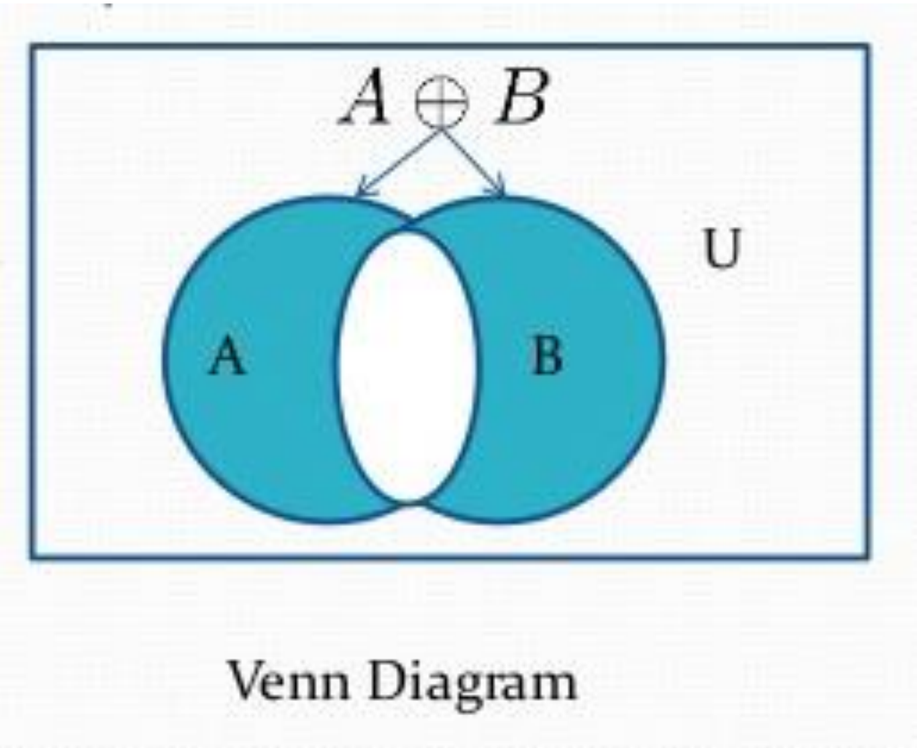
$A - B = \{ 1, 3, 5, 7 \}$

Venn diagram of *A-B*



Symmetric Difference

The symmetric difference of set *A* and set *B*, denoted by $A \oplus B$ is the set **(*A* - *B*) U (*B* - *A*)**



Venn Diagram

Example

Let, **U**, **A** and **B** be sets.

$$\mathbf{U} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$\mathbf{A} = \{1, 2, 3, 4, 5\}; \quad \mathbf{B} = \{4, 5, 6, 7, 8\}$$

$$\mathbf{A} \otimes \mathbf{B} = (\mathbf{A} - \mathbf{B}) \cup (\mathbf{B} - \mathbf{A}) = \{1, 2, 3, 6, 7, 8\}$$

$(\mathbf{A} - \mathbf{B}) = \{1, 2, 3\}$

$(\mathbf{B} - \mathbf{A}) = \{6, 7, 8\}$

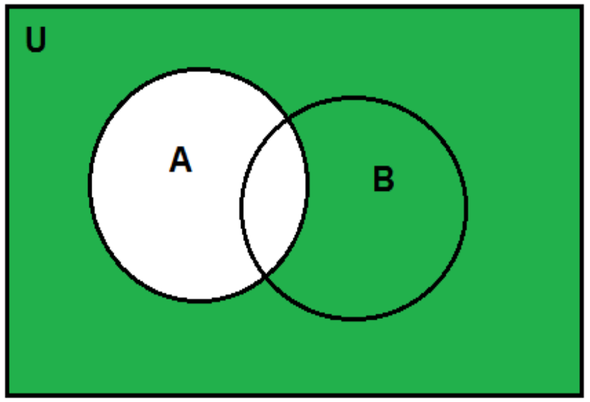
Complement

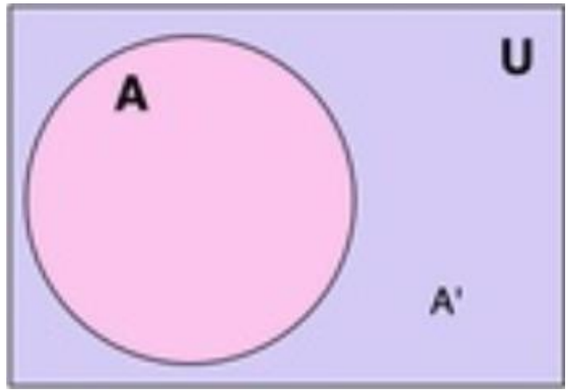
The complement of a set A with respect to a universal set U, denoted by A' is defined to be

$$A' = \{x \in U \mid x \notin A\}$$

$$A' = U - A$$

Venn diagram of *A'*





Example

Let U be a universal set,

$U = \{ 1, 2, 3, 4, 5, 6, 7 \}$
$A = \{ 2, 4, 6 \}$

$A' = U - A = \{ 1, 3, 5, 7 \}$

Set Identities (Properties of Set)

- **Commutative laws**

  $A \cap B = B \cap A, \quad A \cup B = B \cup A$

- **Associative laws**

  $A \cap (B \cap C) = (A \cap B) \cap C$

  $A \cup (B \cup C) = (A \cup B) \cup C$

- **Distributive laws**

  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Set Identities (Properties of Set)

- **Absorption laws**

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

- **Idempotent laws**

$$A \cap A = A, \quad A \cup A = A$$

- **De Morgan's laws**

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Set Identities (Properties of Set)

- **Complement laws**

$$A \cap A' = \emptyset \quad A \cup A' = U$$

- **Double complement laws**

$$(A')' = A$$

- **Complement of *U* and ∅**

$$\emptyset' = U \quad U' = \emptyset$$

Set Identities (Properties of Set)

- **Properties of universal set**

  $A \cup U = U \quad A \cap U = A$

- **Set difference laws**

  $A - B = A \cap B'$

- **Identity laws**

  $A \cup \emptyset = A \quad A \cap U = A$

- **Properties of empty set**

  $A \cup \emptyset = A \quad A \cap \emptyset = \emptyset$

Example

- Let A, B and C denote the subsets of a set S and let C' denote a complement of C in S.
- If $A \cap C = B \cap C$ and $A \cap C' = B \cap C'$, then prove that $A = B$

$$
\begin{aligned}
A &= A \cap S \\
&= A \cap (C \cup C') \\
&= (A \cap C) \cup (A \cap C') && \textbf{(distributive)} \\
&= (B \cap C) \cup (B \cap C') && \textbf{(the given conditions)} \\
&= B \cap (C \cup C') && \textbf{(distributive)} \\
&= B \cap S \\
&= B
\end{aligned}
$$

Example

By referring to the properties of set operations, show that:

$$A - (A \cap B) = A - B$$

set difference
$A - B = A \cap B'$

$A - (A \cap B) = A \cap (A \cap B)'$ [set difference laws]

$= A \cap (A' \cup B')$ [De Morgan's laws]

$= (A \cap A') \cup (A \cap B')$ [distributive laws]

$= \emptyset \cup (A \cap B')$ [complement laws]

$= (A \cap B') \cup \emptyset$ [commutative]

$= A \cap B'$ [Identity laws]

$= A - B$

Generalized Unions and Intersections

The ***union*** of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

Notation:

$$\bigcup_{i=0}^{n} A_i = \{x \in \mathbf{U} | x \in A_i \text{ for at least one } i = 0, 1, 2, \ldots, n\}$$

$$\bigcup_{i=0}^{\infty} A_i = \{x \in U | x \in A_i \text{ for at least one nonnegative integer } i\}$$

Generalized Unions and Intersections

The ***intersection*** of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

Notation:

$$\bigcap_{i=0}^{n} A_i = \{x \in \mathbf{U} | x \in A_i \text{ for all } i = 0, 1, 2, \ldots, n\}$$

$$\bigcap_{i=0}^{\infty} A_i = \{x \in \mathbf{U} | x \in A_i \text{ for all nonnegative integer } i\}$$

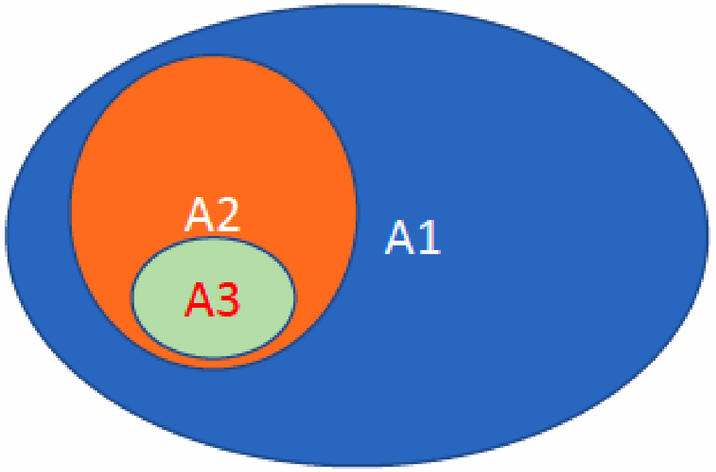
Example

For each positive integer i, let $A_i = \left\{x \in \mathbf{R} \mid -\frac{1}{i} < x < \frac{1}{i}\right\} = A_i = \left(-\frac{1}{i}, \frac{1}{i}\right)$.

Find:

a) $A_1 \cup A_2 \cup A_3$

b) $A_1 \cap A_2 \cap A_3$



Solution (a):

a) $A_1 = \{-1,1\}$; $A_2 = \left\{-\frac{1}{2}, \frac{1}{2}\right\}$; $A_3 = \left\{-\frac{1}{3}, \frac{1}{3}\right\}$

$\therefore A_1 \cup A_2 \cup A_3 = \left\{x \in \mathbf{R} \mid x \text{ is in at least one of the interval } (-1,1) \text{ or } \left(-\frac{1}{2}, \frac{1}{2}\right) \text{ or } \left(-\frac{1}{3}, \frac{1}{3}\right)\right\}$

$= \{x \in \mathbf{R} \mid -1 < x < 1\}$

$= (-1,1)$

Because all the elements in $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $\left(-\frac{1}{3}, \frac{1}{3}\right)$ are in $(-1,1)$

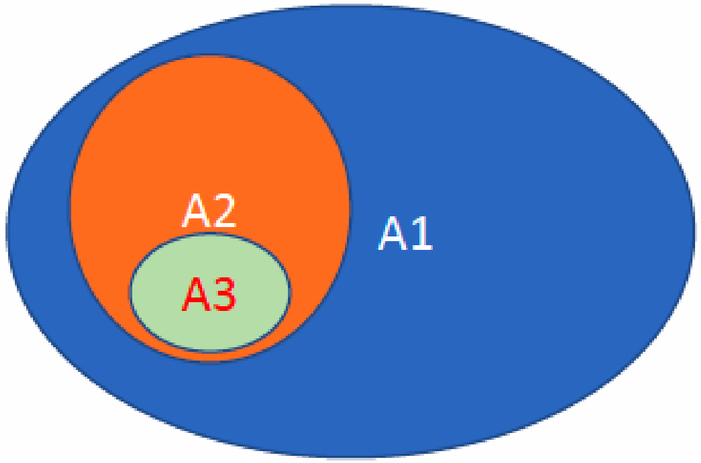
Example

For each positive integer i, let $A_i = \left\{x \in \mathbf{R} \mid -\frac{1}{i} < x < \frac{1}{i}\right\} = A_i = \left(-\frac{1}{i}, \frac{1}{i}\right)$.

Find:

a) $A_1 \cup A_2 \cup A_3$

b) $A_1 \cap A_2 \cap A_3$



Solution (b):

a) $A_1 = \{-1,1\}$; $A_2 = \left\{-\frac{1}{2}, \frac{1}{2}\right\}$; $A_3 = \left\{-\frac{1}{3}, \frac{1}{3}\right\}$

$\therefore A_1 \cap A_2 \cap A_3 = \left\{x \in \mathbf{R} \mid x \text{ is in all of the interval } (-1,1) \text{ and } \left(-\frac{1}{2}, \frac{1}{2}\right) \text{ and } \left(-\frac{1}{3}, \frac{1}{3}\right)\right\}$

$= \left\{x \in \mathbf{R} \mid -\frac{1}{3} < x < \frac{1}{3}\right\}$

$= \left(-\frac{1}{3}, \frac{1}{3}\right)$

Because $\left(-\frac{1}{3}, \frac{1}{3}\right) \subseteq \left(-\frac{1}{2}, \frac{1}{2}\right) \subseteq (-1,1)$

Cartesian Product

- Let A and B be sets. An ordered pair of elements $a \in A$ dan $b \in B$ written (a, b) is a listing of the elements a and b in a specific order.

- The ordered pair (a, b) specifies that a is the first element and b is the second element.

- An ordered pair (a, b) is considered distinct from ordered pair (b, a), unless $a=b$.

Example $(1, 2) \neq (2, 1)$

Cartesian Product

- The Cartesian product of two sets A and B, written $A \times B$ is the set,

$$A \times B = \{(a,b) | \ a \in A, \ b \in B\}$$

- For any set A,

$$A \times \emptyset = \emptyset \times A = \emptyset$$

Example

$A = \{a, b\}$, $B = \{1, 2\}$.

$A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$

$B \times A = \{(1, a), (1, b), (2, a), (2, b)\}$

Cartesian Product

- if $A \neq B$, then $A \times B \neq B \times A$.
- if $|A| = m$ and $|B| = n$, then $|A \times B| = mn$.

Example $A = \{1, 3\}$, $B = \{2, 4, 6\}$.

$A \times B = \{(1, 2), (1, 4), (1, 6), (3, 2), (3, 4), (3, 6)\}$

$B \times A = \{(2, 1), (2, 3), (4, 1), (4, 3), (6, 1), (6, 3)\}$

$A \neq B$, $A \times B \neq B \times A$

$|A| = 2$, $|B| = 3$, $| A \times B | = 2.3 = 6$.

Cartesian Product

- The Cartesian product of sets $A_1, A_2, \ldots, A_n$ is defined to be the set of all n-tuples

  $(a_1, a_2, \ldots a_n)$ where $a_i \in A_i$ for $i=1,\ldots,n$;

- It is denoted $A_1 \times A_2 \times \ldots \times A_n$

  $|A_1 \times A_2 \times \ldots \times A_n| = |A_1| . |A_2| \ldots |A_n|$

Example

$A = \{a, b\}$, $B = \{1, 2\}$, $C = \{x, y\}$

$A \times B \times C = \{(a,1,x),(a,1,y), (a,2,x), (a,2,y), (b,1,x), (b,1,y), (b,2,x), (b,2,y)\}$

$|A \times B \times C| = 2 \times 2 \times 2 = 8$

Additional example: Student skill sets

$U = \{Ali, Mei, Ravi, Siti, Kumar\}$

$P = \{Ali, Mei, Ravi\}$ $S = \{Mei, Ravi, Siti\}$

$P \cup S = \{Ali, Mei, Ravi, Siti\}$ $P \cap S = \{Mei, Ravi\}$

$P - S = \{Ali\}$ $P \oplus S = \{Ali, Siti\}$ $P' = \{Siti, Kumar\}$

$|P \cup S| = 3 + 3 - 2 = 4$

Clarification: For finite sets, $|A \times B| = |A|\,|B|$. Different sets can still have equal Cartesian products when one set is empty.

Thank You